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Life Outside this World: Sustained Research Paper

Introduction/ Background

Life existing outside of our current home seems impossible to many people. The idea of alien creatures is a fun thought but most just leave it at that, a fun thought. Life outside this world must be possible though. Scientists would have to have very closed minds to believe that Earth is the only place in the entire universe that can hold life. Science is all about curiosity and discovery. This research paper will be focusing on our solar system and life closer to home. There are many moons in our solar system, and many have prospects for life. Unicellular life, or life that contains one cell, could be just below our noses, and we would never notice unless we took a closer look. Scientists are more likely to find life on moons in our galaxy in the near future than on planets due to the high life potential of many moons in our solar system.

Life is created through a process called Abiogenesis. Abiogenesis is the formation of life from an inorganic source (Cole 173). This formation takes hold when researchers discuss the beginning of life on Earth as well as outside our planet. Minute life, or simple life, can be formed from inorganic sources and grow to become multicellular organisms over time. To create this evolution, an environment needs basic life-giving resources such as food and liquid water (Cole 172). Basic needs for the creation of life are still being debated among scientists, but certain features will always remain the same, such as liquid water.

One of the creatures here on Earth that scientists look at to discover life-giving properties is nematodes. Nematodes or worms come in all shapes and sizes all around the globe. The worms most studied by scientists are the super tiny ones. These tiny nematodes can live deep underground where there is no air, they can also survive travel to space and in the International Space Station. There were even some discovered 1.3 kilometers underground (Borgonie e et al. 79). Scientists look to Earth for many examples and comparisons to planets or moons to find where there may be life.

Argument/ Rebuttal

One of the reasons it is so hard for life to evolve in space is that it is a desolate void. Life is an incredible and strong force, but not even the toughest animals can survive in certain environments. In order for life to cultivate we need to go back to the basic biology of life, "In a search for extraterrestrial life in our solar system, … a less ambitious notion of 'life as we know it,' meaning life-based on a liquid water solvent, a suite of ''biogenic'' elements … and a source of free energy (7)." (Chyba and Phillips 801). Liquid water tends to be the main idea that scientists describe as a sign of life. So far humans have not discovered liquid water on other planets, but they have discovered signs that there was running water in the past. Scientists have also found a lot of ice on other planets in our solar system. Other than water, life needs heat. While some life can survive in the coldest and darkest places there is a degree to which life can no longer be sustained. The environment around any source of life needs to be warm or cold enough for liquid water to survive, especially at the beginning of life. The first life here on Earth for example was formed in the ocean. Having a source of heat and water for food is enough to cultivate and sustain life. (Chela-Flores 102)

Why Moons?

The Planets in our solar system may seem like the right place to look for life as we live on a planet, not a moon, but they all have different intriguing properties that make them uninhabitable. Starting with the planets closest to the sun most are far too hot to sustain life let alone liquid water. Venus, our sister planet may seem more capable as it is in a better temperature range, but unfortunately, it has an atmosphere so dense that the heat from the spinning orb cannot escape and is trapped in a dense chemical storm. Past us is Mars, another good candidate for life, but Mars has no atmosphere and therefore no heat regulation for life to form. Mars is very cold during the night and warm during the day leaving two extremes. Past Mars are all Jovian planets, made with materials that are light like air. Most of these planets are made of gas and dust making it impossible for life to form or spacecraft to land on them. This leads the search for life to moons. Many planets have moons terrestrial and Jovian alike. Jovian planets tend to be magnets for moons though. (Chela-Flores 101)

The moons in our solar system have a high potential for life. Each moon is a diverse and different environment. Moons are satellites of other planets which means they orbit around the planets instead of the sun. Any natural satellite can be a moon, “Regardless of size, any object that orbits a planet, dwarf planet, or asteroid is a moon.” (Miller 17). The main moons scientists are looking at for life orbit Jupiter. These moons have been highly looked at for life in our solar system, but we have yet to touch down on these moons. Scientists study these moons through flybys and telescopes. Jupiter is about four to six astronomical units or suns away from us (Miller 29–31). That distance alone would take years to cover so it is very hard to send spacecraft out to these moons. Scientists must rely on their observations when studying these moons instead of hard evidence such as soil samples.

We can see from Earth however some large life-giving properties on the moons. While these moons are very far from the sun for instance they are still able to maintain a decent temperature due to their atmospheres. Moons can have atmospheres just like planets and similar to Earth these atmospheres protect the moon from many things. Looking at the age of the geology on a planet helps scientists to determine if a moon has an atmosphere or not. If astronomers observe that a moon has close to no craters, then it may be protected by an atmosphere or there may be active volcanoes on the moon which can create an atmosphere and the warmth that cultivates life (Miller 29–31).

Scientists and astronomers can also observe the orbits and gravitational pull of these moons from their planet. Jupiter is a massive planet and therefore has a strong gravitational force on these moons. The speed of an orbit and the speed of a rotation can change the climate of a moon creating various wind patterns and even Geographical patterns. Similar to Earth and its moon Jupiter's moons feel the pull of the tide as they orbit. On Earth, we observe this as the water rising and falling based on the position of the moon but really the entire planet is being stretched. This stretching of a planet or moon can cause eruptions squeezing magma out of the planet or moon. The stretching of the moons from Jupiter can heat up the ground through friction creating a warmer and more inhabitable environment. (Miller 33–34)

One of the moons Astronomers look at for life is Ganymede (Chela-Flores 102). Ganymede is the largest moon in our solar system, and it orbits the Jovian planet Jupiter. Ganymede has an iron core similar to Earth and is bigger than the planet Mercury. Its surface consists of rock and ice, this ice flows over the moon like glaciers do on Earth. Ganymede also has odd cracks in its surface depicting that there may be some geographical activity still occurring (Miller 37–38). The main evidence for life on Ganymede is the Atmosphere and ice on the surface. Through spectral detection, scientists have seen that Ganymede has certain organic substances in its atmosphere (Chyba and Phillips 801). There are expected to be liquid ocean under the surface of Ganymede and these oceans could cultivate unicellular life (Chela-Flores 103). The main focus of Astronomers is liquid oceans on the moons. Due to the beginnings of life on Earth Scientists believe that liquid oceans are our current best sign of life in the universe.

The other moon highly looked at by scientists is Europa. Europa is only slightly smaller than Earth and covered in ice. The icy surface of Europa is very smooth indicating that the surface is very young. This young surface shows us that it is still changing, and the ice does not stay completely frozen. From this young surface, scientists have concluded that there must be a subsurface ocean on Europa. This discovery creates great strides for the argument that life may exist on Europa (Miller 38). If there is an ocean then multicellular life may even exist, “Our current understanding … of Europa provide good reason to expect that Europa may be habitable at present and may have been habitable throughout much of its past.” (Pappalardo et al. 621). Europa has almost twice as much water as Earth, further backing up this argument for life. There are also geysers on Europa that spout water through the icy sheet and onto the surface keeping it smooth (Miller 39). These geysers could be the result of the pull from Jupiter indicating again that this ocean could contain life. In Earth's oceans, life lives deep beneath the surface using underwater volcanoes and geysers for warmth, “Europa’s putative subsurface ocean suggests that the traditional view of planetary habitability should be broadened (7, 11, 18).” (Chyba and Phillips 801). Scientists can infer that Europa has similar seafloor volcanoes and geysers creating warmth for creatures to grow. The idea itself of sea creatures living on a different celestial body is amazing.

Conclusion

To further prove there may be life on moons such as Europa we need to study Earth more. Only a small portion of the ocean has been researched and there is still a lot to learn about the development of life. Scientists need to come together in this adventure to discover life. Physicists, astronomers, and biologists could change the course of science with new discoveries. One group of scientists stated “Europa may be the premier place in the solar system to search for both extant life and a second origin of life. The discovery of extant life beyond Earth would provide critical information about the fundamental nature of life and the breadth of possible biological processes.”(Pappalardo et al. 587). This statement has given many scientists the urge to continue their research no matter how difficult.

These moons further help the argument that astronomers are most likely to find life on moons in the solar system than on planets. The planets of the solar system are desolate and lack the properties needed for abiogenesis. Even if a planet could support some life introduced to the environment it would not last long in the new environment, “Rather than facing the complex problem of the origin of life on Earth, by means of the time-honored reductionist methods of chemical evolution, the main thesis of the present work is instead to focus on the origin of habitable ecosystems on Earth and elsewhere.”(Chela-Flores 102). Abiogenesis must occur for new life to evolve for the specific planet (Hand). Europa and Ganymede hold these life-giving properties giving them the best chance at developing and supporting life.

Together Ganymede and Europa prove that there are life-giving properties on moons and that moons are scientists' best chance at finding life in our solar system. Europa's subsurface oceans could even have multicellular life which could create great strides in science. The search for these habitable ecosystems has been a long one and will continue for years to come. The planets in our solar system are not fit to create or sustain life, but some moons are. Moons contain many life-giving properties that could lead to life or already have.

Works Cited

Borgonie, G., et al. “Nematoda from the Terrestrial Deep Subsurface of South Africa.” *Nature*, vol. 474, no. 7349, 7349, June 2011, pp. 79–82. *www.nature.com*, https://doi.org/10.1038/nature09974.

Chela-Flores, J. *Instrumentation for the Search for Habitable Ecosystems in the Future Exploration of Europa and Ganymede*. no. 2, Apr. 2010, pp. 101–08. *Cambridge University Press*, https://doi.org/10.1017/S1473550410000029.

Chyba, Christopher F., and Cynthia B. Phillips. “Possible Ecosystems and the Search for Life on Europa.” *Proceedings of the National Academy of Sciences*, vol. 98, no. 3, Jan. 2001, pp. 801–04. *pnas.org (Atypon)*, https://doi.org/10.1073/pnas.98.3.801.

Cole, George H. A. *Inverted Bowl, The: Introductory Accounts Of The Universe And Its Life*. World Scientific Publishing Company, 2009. *ProQuest Ebook Central*, http://ebookcentral.proquest.com/lib/vt/detail.action?docID=731245.

Hand, Kevin. Astrobiology and Potential for Life on Europa. https://drive.google.com/drive/folders/1wrrby0iEVnHXPOchOSIiYIruwcHIPoV6. Accessed 5 Apr. 2023

Miller, Ron. *Natural Satellites: The Book of Moons*. Lerner Publishing Group, 2021. *ProQuest Ebook Central*, http://ebookcentral.proquest.com/lib/vt/detail.action?docID=6416877.

Pappalardo, Robert T., et al., editors. *Europa*. University of Arizona Press, 2017. *DOI.org (Crossref)*, https://doi.org/10.2307/j.ctt1xp3wdw.